SYSTEM ADMINISTRATION & IT INFRASTRUCTURE

Practical Scenario-Based Assignment

Course	System Administration & IT Infrastructure Management
Total Marks	100 Marks
Submission Format	MS Word (.docx) or PDF — with annotated screenshots
Student Name	________________________________
Student ID	________________________________

GENERAL INSTRUCTIONS

- Read each scenario carefully before attempting the task.
- All practical tasks require annotated screenshots as evidence.
- Write observations, commands, and analysis in your own words.
- Partial marks are awarded — do not leave any task blank.
- Submit in MS Word (.docx) or PDF format only.
- Academic integrity policy applies — collaboration is permitted, copying is not.

Marks Distribution Summary

Q#	Topic / Scenario	Marks	CLO
Q1	Lab Setup & Windows Server Configuration	15	CLO-1
Q2	Linux Terminal Administration	15	CLO-2
Q3	PowerShell Automation Scripting	20	CLO-3
Q4	Active Directory Domain Services	25	CLO-4
Q5	Ethics & Professional Conduct	15	CLO-5
Q6	Career Reflection & Growth Mindset	10	CLO-6
	TOTAL	**100**	

Question 1: Lab Setup & Windows Server Configuration (15 Marks)

Scenario: You have joined a small IT consultancy as a junior system administrator. Your manager has asked you to set up a basic lab environment to test Windows Server 2019 before deploying it to a client site.

Task 1.1 — Virtual Lab Setup (5 Marks)

Using VirtualBox or VMware, set up a virtual machine with Windows Server 2019 installed. Once the installation is complete:

- Configure a Static IP address for the server.
- Set up a Virtual Switch to isolate the lab network from your host machine.
- Verify connectivity by pinging the server from the host.

Evidence Required:

- Screenshot of the VM running with network adapter settings visible.
- Screenshot of the static IP configuration (ipconfig /all output).
- Screenshot of a successful ping result.

Task 1.2 — Windows Admin Center Exploration (5 Marks)

Install and open Windows Admin Center on your server. Navigate through the management interface and answer the following:

- What three management categories can you see in the Windows Admin Center dashboard? List them.
- How does Windows Admin Center differ from the legacy Microsoft Management Console (MMC)? Write 3 to 4 lines comparing both.

Evidence Required:

- Screenshot of the Windows Admin Center dashboard with your server connected.

Task 1.3 — Server Innovation Identification (5 Marks)

Based on your installation and exploration, identify TWO modern features specific to Windows Server 2019 or 2022. For each feature:

- State the feature name.
- Explain in 2 to 3 sentences how it improves enterprise administration compared to older versions.

Question 2: Linux Terminal Administration (15 Marks)

Scenario: A client's Ubuntu server has been handed to you for routine maintenance. You must complete several terminal-based tasks to verify the system's health and file structure.

Task 2.1 — File System Navigation (4 Marks)

Open a terminal and perform the following steps in sequence. Capture screenshots at each step:

1. Display your current directory using the appropriate command.
2. Navigate to the /etc directory and list its contents in long format showing file permissions.
3. Create a new directory called lab_work inside your home folder.
4. Inside lab_work, create a text file named server_notes.txt and add two lines of text to it.

Evidence Required:

- Annotated screenshots showing each command and its output.

Task 2.2 — Permissions Management (6 Marks)

On the server_notes.txt file you created:

- Check and display the current permissions of the file.
- Modify permissions so that: the owner has read and write access, the group has read-only access, and others have no access. Use the chmod command with numeric notation.
- Change the file's group ownership to a new group called admin_team (create the group first if it does not exist).

After completing the above, explain in 3 to 4 sentences: why is the Principle of Least Privilege important when assigning Linux file permissions? Provide a real-world example.

Evidence Required:

- Screenshot of the ls -l output before and after permission changes.
- Screenshot showing chown/chgrp commands executed.

Task 2.3 — Text Processing (5 Marks)

Create a file called users.txt with the following fictional content:

```
alice:admin:active bob:user:inactive carlos:admin:active diana:user:active
```

Use terminal commands to:

- Display only the lines where the role is admin.
- Count the number of active users.
- Replace the word inactive with suspended in the file using sed, and display the updated file.

Evidence Required:

- Screenshot of each command and its output.

Question 3: PowerShell Automation Scripting (20 Marks)

Scenario: Your organization manages 50 Windows servers. Your manager has asked you to write PowerShell scripts to automate routine monitoring and reporting tasks instead of checking each server manually.

Task 3.1 — Service Health Check Script (7 Marks)

Write a PowerShell script that:

- Retrieves all running services on the local machine.
- Filters and displays only those services whose display name contains the word Windows.
- Outputs the result in a formatted table showing Name, Status, and StartType.
- Uses a Try/Catch block to handle any errors gracefully, displaying a user-friendly message if a failure occurs.

Include the script code in your document and explain what each major section does in plain English (3 to 4 sentences total).

Evidence Required:

- Full script code pasted into your document.
- Screenshot of the script running in PowerShell ISE or VS Code and showing output.

Task 3.2 — Automated Reporting with Loops (8 Marks)

Write a PowerShell script that generates a basic system report. The script must:

- Use a ForEach loop to iterate through a predefined list of at least three computer names (can be the local machine name repeated for testing).
- For each computer, retrieve: Computer Name, Operating System, Last Boot Time.
- Append the results to a text file called system_report.txt in a readable format.
- Display a confirmation message after each entry is written.

After running the script, open system_report.txt and include its contents in your submission.

Evidence Required:

- Full script code in the document.
- Screenshot of the script execution.
- Contents of system_report.txt pasted into the document.

Task 3.3 — Remoting Concept (5 Marks)

Without necessarily running the command (as a lab environment may not have multiple machines), write the PowerShell command that would:

- Run the Get-Service command on three remote machines simultaneously using Invoke-Command.

Then answer the following questions in 2 to 3 sentences each:

- What execution policy would you apply before running scripts in a production environment, and why?
- What is the difference between One-to-One and One-to-Many remoting in PowerShell?

Question 4: Active Directory Domain Services (25 Marks)

Scenario: TechVenture Ltd. is a growing company with 3 departments: HR, Finance, and IT. You have been hired to build and manage their Active Directory infrastructure from scratch using Windows Server 2019.

Task 4.1 — Domain Setup & OU Structure (8 Marks)

On your Windows Server 2019 VM, promote the server to a Domain Controller and create the domain techventure.local. Once the domain is set up:

- Create three Organizational Units (OUs): HR, Finance, and IT.
- Inside each OU, create at least two user accounts with realistic names.
- Create one Security Group per department (e.g., HR_Staff, Finance_Staff, IT_Staff) and add the relevant users.

In your own words, explain in 3 to 4 sentences: why are OUs used instead of just placing all users in the default Users container?

Evidence Required:

- Screenshot of Active Directory Users and Computers (ADUC) showing the OU tree.
- Screenshot showing at least one user account inside an OU.
- Screenshot of one Security Group with members listed.

Task 4.2 — Additional Domain Controller (ADC) (8 Marks)

Set up a second virtual machine and join it to the techventure.local domain. Promote it to an Additional Domain Controller (ADC).

- Verify AD replication between the primary DC and the ADC using the repadmin /showrepl command.
- Simulate a failure scenario: shut down the primary DC and confirm that domain users can still log in via the ADC.

Answer the following in 4 to 5 sentences: what is the purpose of deploying an Additional Domain Controller, and how does it support fault tolerance and load balancing in an enterprise environment?

Evidence Required:

- Screenshot of the repadmin /showrepl output showing successful replication.
- Screenshot showing a domain login from the workstation while the primary DC is offline.

Task 4.3 — Domain Join a Workstation (5 Marks)

Set up a Windows 10 or Windows 11 virtual machine (or use an existing one). Join it to the techventure.local domain:

- Before joining, verify that the workstation's DNS is pointing to the Domain Controller's IP.

- Complete the domain join process and restart the machine.
- Log in to the workstation using a domain user account created in Task 4.1.

Evidence Required:

- Screenshot of the DNS settings on the workstation pointing to the DC.
- Screenshot of the System Properties dialog confirming domain membership.
- Screenshot of a successful domain login on the workstation.

Task 4.4 — User vs Group Differentiation (4 Marks)

In a short written response (3 to 5 sentences each), answer:

- What is the difference between a Security Group and a Distribution Group in Active Directory? In which scenario would you use each?
- TechVenture Ltd. needs to send a company-wide email and also restrict access to a shared Finance folder. Which group type would you use for each purpose and why?

Question 5: Ethics & Professional Standards (15 Marks)

Scenario: You are a senior system administrator at a healthcare organization. You have access to patient databases, employee payroll systems, and the company's entire network infrastructure. Your decisions directly affect data security and people's livelihoods.

Task 5.1 — Ethical Dilemma Analysis (7 Marks)

Read the following case study and answer the questions below:

> *Case Study: Your colleague, a junior admin, accidentally deleted a critical database backup. They come to you privately and ask you to cover it up by backdating the system logs so no one finds out. They argue that the data can be recovered manually and no real harm has been done.*

Answer the following questions:

5. Identify which core ethical principle(s) from your course content are being violated in this scenario. Justify your answer. (3 marks)
6. What would you do in this situation? Provide a step-by-step explanation of your response, referencing professional standards you have learned. (4 marks)

Task 5.2 — Professional Best Practices Application (8 Marks)

As the senior administrator of the healthcare organization, describe how you would implement the following in your day-to-day operations. For each, write 3 to 5 sentences explaining the practice and providing a specific example relevant to a healthcare IT environment:

a) Principle of Least Privilege

Describe how you would apply this principle when setting up user accounts for doctors, nurses, and administrative staff. What access would each role have and what would be restricted?

b) Change Management Discipline

Explain the steps you would follow before making any changes to the production server. Why is documentation and approval important in a healthcare environment specifically?

c) Blameless Postmortem Culture

Describe how you would conduct a postmortem meeting after the database backup incident from Task 5.1. What questions would you ask, and what outcomes would you aim for?

Question 6: Career Reflection & Growth Mindset (10 Marks)

Scenario: You are preparing for a job interview for a Cloud Systems Engineer role. The interviewer asks you to reflect on your learning journey, technical growth, and career aspirations in system administration.

Task 6.1 — Historical Evolution Reflection (4 Marks)

In 1 paragraph (5 to 7 sentences), describe the evolution of system administration from the Mainframe Era to today's Hybrid Cloud and AI-driven landscape. In your reflection:

- Identify at least TWO major turning points in this evolution.
- Explain how each turning point changed the role and responsibilities of a system administrator.
- Reflect on which era you would have found most challenging to work in, and why.

Task 6.2 — Specialization Research & Career Planning (6 Marks)

Choose ONE of the following high-value specializations mentioned in your course:

Cloud Architect	DevOps Engineer	Site Reliability Engineer

For your chosen specialization, research and write the following (note: you may use external sources but cite them briefly):

7. Describe what this role does day-to-day in a large organization. (2 marks)
8. List THREE technical skills or certifications most commonly required for this role. (2 marks)
9. Explain in 3 to 5 sentences how a growth mindset — as discussed in class — is particularly important for someone in this role. Give a specific example of a challenge they might face and how a growth mindset would help them overcome it. (2 marks)

Marking Rubric — All Questions

Criterion	Excellent (90–100%)	Good (70–89%)	Satisfactory (50–69%)	Needs Improvement (<50%)
Task Completion	All steps completed accurately with no gaps	Most steps done, minor omissions	Some steps attempted, significant gaps	Incomplete or largely incorrect
Screenshot Evidence	Clear, annotated, directly relevant screenshots provided	Most screenshots present, minor annotation missing	Some screenshots, unclear or irrelevant	Missing or no screenshots

Written Explanation	Accurate, detailed, uses correct technical terminology	Mostly accurate, adequate detail	Partially correct, limited detail	Incorrect or missing explanations
Ethics & Professionalism	Demonstrates strong ethical reasoning with real-world application	Good ethical reasoning, some application	Basic ethical understanding shown	Limited or no ethical analysis

— END OF ASSIGNMENT —